

# prode-ML

## FIFA World Cup 2026 - Predictions Report

*Generated: 2026-06-04 09:27*

### Model Metrics:

Training date: 2026-06-04T09:27:38

Samples: 3199 matches, 4478 train

Date range: 2000-01-09 to 2026-06-03

Split: time\_series\_chronological, test from 2022-06-10

#### Model Accuracies (test set):

|                                            |       |
|--------------------------------------------|-------|
| RandomForest                               | 49.0% |
| XGBoost                                    | 46.7% |
| LightGBM                                   | 49.6% |
| CatBoost                                   | 51.0% |
| Weighted voting                            | 50.2% |
| Two-Stage                                  | 35.4% |
| Confederation                              | 48.3% |
| High Elo diff ( $\geq 200$ ): 57.8% (n=71) |       |
| Baseline random: 33.3%                     |       |

# 1. Group Stage Predictions

All 72 group matches with predicted probabilities and expected goals.

## GROUP A: Mexico, South Africa, South Korea, Czechia

| Match                      | Win A | Draw  | Win B | xG      | Most Likely                                                                           | Conf | Upset |
|----------------------------|-------|-------|-------|---------|---------------------------------------------------------------------------------------|------|-------|
| Mexico v South Africa      | 63.6% | 21.4% | 15.0% | 1.1-0.8 | Mexico vs South Africa -- error: tuple indices must be integers or strings, not tuple |      |       |
| Mexico v South Korea       | 40.3% | 30.8% | 28.9% | 1.2-0.9 |                                                                                       |      |       |
| Mexico v Czechia           | 42.6% | 25.2% | 32.2% | 1.1-1.2 |                                                                                       |      |       |
| South Africa v South Korea | 18.6% | 22.1% | 59.3% | 0.8-0.9 |                                                                                       |      |       |
| South Africa v Czechia     | 21.4% | 23.5% | 55.1% | 0.7-1.2 |                                                                                       |      |       |
| South Korea v Czechia      | 36.1% | 27.2% | 36.6% | 0.9-1.3 |                                                                                       |      |       |

## GROUP B: Canada, Switzerland, Qatar, Bosnia and Herzegovina

| Match                                | Win A | Draw  | Win B | xG      | Most Likely                                                                          | Conf | Upset |
|--------------------------------------|-------|-------|-------|---------|--------------------------------------------------------------------------------------|------|-------|
| Canada v Switzerland                 | 19.5% | 22.0% | 58.5% | 0.5-0.8 | Canada vs Switzerland -- error: tuple indices must be integers or strings, not tuple |      |       |
| Canada v Qatar                       | 48.9% | 27.2% | 23.9% | 0.5-0.7 |                                                                                      |      |       |
| Canada v Bosnia and Herzegovina      | 43.0% | 30.6% | 26.5% | 0.5-0.7 |                                                                                      |      |       |
| Switzerland v Qatar                  | 57.9% | 25.1% | 17.0% | 0.8-0.7 |                                                                                      |      |       |
| Switzerland v Bosnia and Herzegovina | 58.5% | 27.6% | 13.9% | 0.7-0.7 |                                                                                      |      |       |
| Qatar v Bosnia and Herzegovina       | 33.9% | 27.4% | 38.7% | 0.6-0.7 |                                                                                      |      |       |

## GROUP C: Brazil, Morocco, Scotland, Haiti

| Match              | Win A | Draw  | Win B | xG      | Most Likely                                                                      | Conf | Upset |
|--------------------|-------|-------|-------|---------|----------------------------------------------------------------------------------|------|-------|
| Brazil v Morocco   | 59.8% | 24.1% | 16.1% | 1.9-1.3 | Brazil vs Morocco -- error: tuple indices must be integers or strings, not tuple |      |       |
| Brazil v Scotland  | 82.7% | 11.1% | 6.1%  | 1.1-0.9 |                                                                                  |      |       |
| Brazil v Haiti     | 84.7% | 9.7%  | 5.6%  | 0.9-0.9 |                                                                                  |      |       |
| Morocco v Scotland | 54.6% | 27.1% | 18.3% | 0.7-0.9 |                                                                                  |      |       |
| Morocco v Haiti    | 71.6% | 18.3% | 10.1% | 0.6-0.9 |                                                                                  |      |       |
| Scotland v Haiti   | 34.8% | 24.4% | 40.8% | 0.4-0.5 |                                                                                  |      |       |

## GROUP D: United States, Paraguay, Australia, Turkiye

| Match                     | Win A | Draw  | Win B | xG      | Most Likely                                                                              | Conf | Upset |
|---------------------------|-------|-------|-------|---------|------------------------------------------------------------------------------------------|------|-------|
| United States v Paraguay  | 41.1% | 26.0% | 32.9% | 1.0-0.8 | United States vs Paraguay -- error: tuple indices must be integers or strings, not tuple |      |       |
| United States v Australia | 40.2% | 26.1% | 33.7% | 1.0-1.1 |                                                                                          |      |       |
| United States v Turkiye   | 32.7% | 28.2% | 39.1% | 1.0-1.0 |                                                                                          |      |       |
| Paraguay v Australia      | 28.0% | 29.1% | 42.9% | 0.7-1.0 |                                                                                          |      |       |
| Paraguay v Turkiye        | 28.1% | 21.8% | 50.1% | 0.7-0.8 |                                                                                          |      |       |
| Australia v Turkiye       | 26.4% | 28.7% | 44.9% | 1.0-0.8 |                                                                                          |      |       |

## GROUP E: Germany, Ecuador, Cote d'Ivoire, Curacao

| Match                   | Win A | Draw  | Win B | xG      | Most Likely                                                                       | Conf | Upset |
|-------------------------|-------|-------|-------|---------|-----------------------------------------------------------------------------------|------|-------|
| Germany v Ecuador       | 56.7% | 21.9% | 21.4% | 1.4-0.8 | Germany vs Ecuador -- error: tuple indices must be integers or strings, not tuple |      |       |
| Germany v Cote d'Ivoire | 71.8% | 17.3% | 10.9% | 1.5-1.1 |                                                                                   |      |       |
| Germany v Curacao       | 68.0% | 19.9% | 12.1% | 1.3-0.9 |                                                                                   |      |       |
| Ecuador v Cote d'Ivoire | 43.2% | 32.9% | 23.9% | 0.7-0.9 |                                                                                   |      |       |
| Ecuador v Curacao       | 52.8% | 28.6% | 18.5% | 0.6-0.7 |                                                                                   |      |       |
| Cote d'Ivoire v Curacao | 41.7% | 31.7% | 26.7% | 0.9-0.8 |                                                                                   |      |       |

## GROUP F: Netherlands, Sweden, Japan, Tunisia

| Match                | Win A | Draw  | Win B | xG      | Most Likely                                                                          | Conf | Upset |
|----------------------|-------|-------|-------|---------|--------------------------------------------------------------------------------------|------|-------|
| Netherlands v Sweden | 57.4% | 24.6% | 18.0% | 1.1-1.2 | Netherlands vs Sweden -- error: tuple indices must be integers or strings, not tuple |      |       |

|                       |       |       |       |         |
|-----------------------|-------|-------|-------|---------|
| Netherlands v Japan   | 46.0% | 30.4% | 23.6% | 1.3-1.3 |
| Netherlands v Tunisia | 66.7% | 19.4% | 13.9% | 1.4-1.1 |
| Sweden v Japan        | 25.7% | 27.3% | 47.0% | 1.0-1.0 |
| Sweden v Tunisia      | 37.7% | 25.8% | 36.5% | 1.2-0.8 |
| Japan v Tunisia       | 50.5% | 26.0% | 23.5% | 1.3-1.0 |

Netherlands vs Japan -- error: tuple indices must be integers or strings, not NoneType

Netherlands vs Tunisia -- error: tuple indices must be integers or strings, not NoneType

Sweden vs Japan -- error: tuple indices must be integers or strings, not NoneType

Sweden vs Tunisia -- error: tuple indices must be integers or strings, not NoneType

Japan vs Tunisia -- error: tuple indices must be integers or strings, not NoneType

## GROUP G: Belgium, Egypt, Iran, New Zealand

| Match                 | Win A | Draw  | Win B | xG      | Most Likely                                                                        | Conf | Upset |
|-----------------------|-------|-------|-------|---------|------------------------------------------------------------------------------------|------|-------|
| Belgium v Egypt       | 61.7% | 19.3% | 18.9% | 1.2-0.8 | Belgium vs Egypt -- error: tuple indices must be integers or strings, not NoneType |      |       |
| Belgium v Iran        | 65.8% | 21.1% | 13.1% | 1.3-0.8 |                                                                                    |      |       |
| Belgium v New Zealand | 78.8% | 13.5% | 7.7%  | 0.7-0.4 |                                                                                    |      |       |
| Egypt v Iran          | 30.6% | 39.1% | 30.3% | 1.2-1.1 |                                                                                    |      |       |
| Egypt v New Zealand   | 57.4% | 26.4% | 16.2% | 0.6-0.6 |                                                                                    |      |       |
| Iran v New Zealand    | 53.8% | 24.7% | 21.5% | 0.6-0.6 |                                                                                    |      |       |

## GROUP H: Spain, Uruguay, Saudi Arabia, Cape Verde

| Match                     | Win A | Draw  | Win B | xG      | Most Likely                                                                        | Conf | Upset |
|---------------------------|-------|-------|-------|---------|------------------------------------------------------------------------------------|------|-------|
| Spain v Uruguay           | 60.8% | 22.7% | 16.4% | 1.4-1.6 | Spain vs Uruguay -- error: tuple indices must be integers or strings, not NoneType |      |       |
| Spain v Saudi Arabia      | 75.6% | 13.8% | 10.6% | 1.2-1.3 |                                                                                    |      |       |
| Spain v Cape Verde        | 82.9% | 10.7% | 6.4%  | 1.2-0.9 |                                                                                    |      |       |
| Uruguay v Saudi Arabia    | 67.4% | 20.7% | 11.9% | 0.9-0.9 |                                                                                    |      |       |
| Uruguay v Cape Verde      | 64.6% | 21.0% | 14.4% | 0.9-0.6 |                                                                                    |      |       |
| Saudi Arabia v Cape Verde | 36.3% | 34.4% | 29.2% | 0.8-0.5 |                                                                                    |      |       |

## GROUP I: France, Senegal, Norway, Iraq

| Match            | Win A | Draw  | Win B | xG      | Most Likely                                                                         | Conf | Upset |
|------------------|-------|-------|-------|---------|-------------------------------------------------------------------------------------|------|-------|
| France v Senegal | 58.0% | 26.3% | 15.7% | 1.4-1.3 | France vs Senegal -- error: tuple indices must be integers or strings, not NoneType |      |       |
| France v Norway  | 68.5% | 17.7% | 13.9% | 1.1-1.2 |                                                                                     |      |       |
| France v Iraq    | 69.6% | 21.7% | 8.7%  | 1.2-1.0 |                                                                                     |      |       |
| Senegal v Norway | 34.6% | 26.5% | 38.9% | 0.8-1.0 |                                                                                     |      |       |
| Senegal v Iraq   | 61.7% | 22.7% | 15.7% | 0.9-0.9 |                                                                                     |      |       |
| Norway v Iraq    | 54.8% | 28.3% | 16.9% | 0.8-0.7 |                                                                                     |      |       |

## GROUP J: Argentina, Austria, Algeria, Jordan

| Match               | Win A | Draw  | Win B | xG      | Most Likely                                                                            | Conf | Upset |
|---------------------|-------|-------|-------|---------|----------------------------------------------------------------------------------------|------|-------|
| Argentina v Austria | 55.8% | 29.1% | 15.1% | 0.9-1.0 | Argentina vs Austria -- error: tuple indices must be integers or strings, not NoneType |      |       |
| Argentina v Algeria | 67.5% | 21.5% | 11.0% | 1.2-1.1 |                                                                                        |      |       |
| Argentina v Jordan  | 76.9% | 15.7% | 7.3%  | 1.2-1.0 |                                                                                        |      |       |
| Austria v Algeria   | 44.1% | 23.6% | 32.3% | 0.7-0.6 |                                                                                        |      |       |
| Austria v Jordan    | 60.9% | 20.4% | 18.7% | 0.8-0.5 |                                                                                        |      |       |
| Algeria v Jordan    | 51.7% | 23.8% | 24.5% | 0.8-0.7 |                                                                                        |      |       |

## GROUP K: Portugal, Colombia, Uzbekistan, DR Congo

| Match                 | Win A | Draw  | Win B | xG      | Most Likely                                                                            | Conf | Upset |
|-----------------------|-------|-------|-------|---------|----------------------------------------------------------------------------------------|------|-------|
| Portugal v Colombia   | 51.6% | 26.1% | 22.3% | 1.4-1.2 | Portugal vs Colombia -- error: tuple indices must be integers or strings, not NoneType |      |       |
| Portugal v Uzbekistan | 70.3% | 17.0% | 12.8% | 1.0-1.1 |                                                                                        |      |       |
| Portugal v DR Congo   | 76.4% | 13.9% | 9.7%  | 0.8-1.1 |                                                                                        |      |       |
| Colombia v Uzbekistan | 67.9% | 17.8% | 14.2% | 0.8-1.1 |                                                                                        |      |       |
| Colombia v DR Congo   | 58.2% | 26.0% | 15.7% | 0.7-1.0 |                                                                                        |      |       |
| Uzbekistan v DR Congo | 30.8% | 34.0% | 35.3% | 0.7-0.8 |                                                                                        |      |       |

## GROUP L: England, Croatia, Ghana, Panama

| Match | Win A | Draw | Win B | xG | Most Likely | Conf | Upset |
|-------|-------|------|-------|----|-------------|------|-------|
|-------|-------|------|-------|----|-------------|------|-------|

|                   |       |       |       |         |
|-------------------|-------|-------|-------|---------|
| England v Croatia | 55.8% | 20.4% | 23.8% | 1.2-1.1 |
| England v Ghana   | 63.5% | 24.7% | 11.8% | 1.3-1.1 |
| England v Panama  | 72.9% | 16.2% | 10.9% | 0.9-0.6 |
| Croatia v Ghana   | 59.0% | 24.8% | 16.2% | 1.1-1.0 |
| Croatia v Panama  | 60.3% | 23.3% | 16.4% | 0.8-0.5 |
| Ghana v Panama    | 39.5% | 22.6% | 37.9% | 0.8-0.6 |

England vs Croatia -- error: tuple indices must be integers

England vs Ghana -- error: tuple indices must be integers

England vs Panama -- error: tuple indices must be integers

Croatia vs Ghana -- error: tuple indices must be integers

Croatia vs Panama -- error: tuple indices must be integers

Ghana vs Panama -- error: tuple indices must be integers

## 2. Group Stage Simulation (100,000 runs)

### GROUP A

| Team         | 1st   | 2nd   | 3rd   | 4th   | Qualify | Avg Pts | Avg GD |
|--------------|-------|-------|-------|-------|---------|---------|--------|
| Czechia      | 36.9% | 27.4% | 20.8% | 15.0% | 64.3%   | 4.8     | 1.02   |
| Mexico       | 30.6% | 27.5% | 23.0% | 18.9% | 58.1%   | 4.4     | 0.45   |
| South Korea  | 18.0% | 23.4% | 27.6% | 31.0% | 41.4%   | 3.6     | -0.58  |
| South Africa | 14.5% | 21.7% | 28.6% | 35.2% | 36.2%   | 3.3     | -0.88  |

**Most Likely Results:**  
Mexico vs South Africa 1-0 (16.2%)  
Mexico vs South Korea 1-0 (13.8%)  
Mexico vs Czechia 1-1 (13.3%)  
South Africa vs South Korea 0-0 (17.5%)  
South Africa vs Czechia 0-1 (17.5%)  
South Korea vs Czechia 0-1 (15.0%)

### GROUP B

| Team               | 1st   | 2nd   | 3rd   | 4th   | Qualify | Avg Pts | Avg GD |
|--------------------|-------|-------|-------|-------|---------|---------|--------|
| Bosnia and Herzego | 30.0% | 26.8% | 23.6% | 19.6% | 56.8%   | 4.3     | 0.46   |
| Switzerland        | 29.1% | 26.2% | 23.5% | 21.2% | 55.3%   | 4.1     | 0.22   |
| Qatar              | 24.5% | 25.0% | 25.4% | 25.0% | 49.5%   | 3.9     | -0.01  |
| Canada             | 16.4% | 22.0% | 27.5% | 34.2% | 38.4%   | 3.2     | -0.67  |

**Most Likely Results:**  
Canada vs Switzerland 0-0 (27.0%)  
Canada vs Qatar 0-0 (29.4%)  
Canada vs Bosnia and Herzegovina 0-0 (31.4%)  
Switzerland vs Qatar 0-0 (22.7%)  
Switzerland vs Bosnia and Herzegovina 0-0 (24.6%)  
Qatar vs Bosnia and Herzegovina 0-0 (26.8%)

### GROUP C

| Team     | 1st   | 2nd   | 3rd   | 4th   | Qualify | Avg Pts | Avg GD |
|----------|-------|-------|-------|-------|---------|---------|--------|
| Brazil   | 38.1% | 25.1% | 19.9% | 16.8% | 63.2%   | 4.6     | 0.84   |
| Haiti    | 26.7% | 28.7% | 26.0% | 18.6% | 55.4%   | 4.3     | 0.44   |
| Scotland | 20.4% | 25.8% | 28.0% | 25.8% | 46.2%   | 3.8     | -0.13  |
| Morocco  | 14.8% | 20.4% | 26.1% | 38.7% | 35.2%   | 3.2     | -1.15  |

**Most Likely Results:**  
Brazil vs Morocco 1-1 (10.0%)  
Brazil vs Scotland 1-0 (14.3%)  
Brazil vs Haiti 0-0 (15.3%)  
Morocco vs Scotland 0-0 (21.4%)  
Morocco vs Haiti 0-0 (24.2%)  
Scotland vs Haiti 0-0 (39.7%)

### GROUP D

| Team          | 1st   | 2nd   | 3rd   | 4th   | Qualify | Avg Pts | Avg GD |
|---------------|-------|-------|-------|-------|---------|---------|--------|
| Australia     | 32.1% | 26.7% | 22.5% | 18.7% | 58.7%   | 4.5     | 0.56   |
| United States | 26.6% | 25.3% | 24.4% | 23.7% | 51.9%   | 4.1     | 0.03   |
| Turkiye       | 24.4% | 25.2% | 25.6% | 24.7% | 49.6%   | 4.0     | 0.02   |
| Paraguay      | 16.9% | 22.8% | 27.5% | 32.8% | 39.7%   | 3.5     | -0.61  |

**Most Likely Results:**  
United States vs Paraguay 0-0 (16.9%)  
United States vs Australia 0-1 (13.6%)  
United States vs Türkiye 1-1 (13.6%)

Paraguay vs Australia 0-0 (19.7%)  
Paraguay vs Türkiye 0-0 (21.4%)  
Australia vs Türkiye 0-0 (16.0%)

GROUP E

| Team          | 1st   | 2nd   | 3rd   | 4th   | Qualify | Avg Pts | Avg GD |
|---------------|-------|-------|-------|-------|---------|---------|--------|
| Germany       | 45.8% | 25.6% | 16.9% | 11.7% | 71.4%   | 5.2     | 1.45   |
| Cote d'Ivoire | 23.1% | 27.1% | 26.0% | 23.8% | 50.2%   | 4.0     | -0.08  |
| Curacao       | 17.6% | 24.9% | 28.5% | 28.9% | 42.5%   | 3.7     | -0.42  |
| Ecuador       | 13.5% | 22.4% | 28.6% | 35.6% | 35.9%   | 3.3     | -0.95  |

**Most Likely Results:**  
Germany vs Ecuador 1-0 (15.6%)  
Germany vs Cote d'Ivoire 1-1 (12.4%)  
Germany vs Curacao 1-0 (14.9%)  
Ecuador vs Cote d'Ivoire 0-0 (18.6%)  
Ecuador vs Curacao 0-0 (25.4%)  
Cote d'Ivoire vs Curacao 0-0 (19.1%)

GROUP F

| Team        | 1st   | 2nd   | 3rd   | 4th   | Qualify | Avg Pts | Avg GD |
|-------------|-------|-------|-------|-------|---------|---------|--------|
| Sweden      | 29.8% | 27.4% | 23.8% | 19.1% | 57.2%   | 4.5     | 0.51   |
| Japan       | 27.4% | 26.2% | 24.8% | 21.6% | 53.6%   | 4.3     | 0.29   |
| Netherlands | 27.5% | 25.8% | 24.3% | 22.5% | 53.2%   | 4.2     | 0.16   |
| Tunisia     | 15.3% | 20.7% | 27.2% | 36.8% | 36.0%   | 3.4     | -0.97  |

**Most Likely Results:**  
Netherlands vs Sweden 1-1 (13.4%)  
Netherlands vs Japan 1-1 (12.9%)  
Netherlands vs Tunisia 1-1 (12.5%)  
Sweden vs Japan 1-0 (14.1%)  
Sweden vs Tunisia 1-0 (16.0%)  
Japan vs Tunisia 1-0 (13.6%)

GROUP G

| Team        | 1st   | 2nd   | 3rd   | 4th   | Qualify | Avg Pts | Avg GD |
|-------------|-------|-------|-------|-------|---------|---------|--------|
| Belgium     | 41.4% | 26.4% | 18.7% | 13.5% | 67.8%   | 4.8     | 1.04   |
| Egypt       | 23.0% | 25.5% | 25.2% | 26.3% | 48.5%   | 3.8     | -0.20  |
| Iran        | 19.7% | 23.6% | 25.5% | 31.2% | 43.3%   | 3.6     | -0.56  |
| New Zealand | 15.8% | 24.5% | 30.6% | 29.0% | 40.4%   | 3.6     | -0.28  |

**Most Likely Results:**  
Belgium vs Egypt 1-0 (15.6%)  
Belgium vs Iran 1-0 (15.1%)  
Belgium vs New Zealand 0-0 (32.7%)  
Egypt vs Iran 1-1 (13.4%)  
Egypt vs New Zealand 0-0 (31.2%)  
Iran vs New Zealand 0-0 (28.7%)

GROUP H

| Team         | 1st   | 2nd   | 3rd   | 4th   | Qualify | Avg Pts | Avg GD |
|--------------|-------|-------|-------|-------|---------|---------|--------|
| Uruguay      | 32.3% | 27.2% | 22.2% | 18.3% | 59.5%   | 4.5     | 0.56   |
| Saudi Arabia | 27.2% | 27.3% | 24.9% | 20.6% | 54.5%   | 4.3     | 0.30   |
| Spain        | 27.5% | 24.9% | 23.6% | 24.0% | 52.4%   | 4.1     | -0.02  |
| Cape Verde   | 12.9% | 20.6% | 29.3% | 37.1% | 33.6%   | 3.2     | -0.85  |

**Most Likely Results:**  
Spain vs Uruguay 1-1 (11.5%)  
Spain vs Saudi Arabia 1-1 (12.8%)  
Spain vs Cape Verde 1-0 (15.2%)

Uruguay vs Saudi Arabia 0-0 (15.8%)  
Uruguay vs Cape Verde 0-0 (21.6%)  
Saudi Arabia vs Cape Verde 0-0 (27.1%)

GROUP I

| Team    | 1st   | 2nd   | 3rd   | 4th   | Qualify | Avg Pts | Avg GD |
|---------|-------|-------|-------|-------|---------|---------|--------|
| France  | 30.4% | 25.1% | 22.6% | 21.9% | 55.5%   | 4.3     | 0.27   |
| Norway  | 28.1% | 26.9% | 24.2% | 20.8% | 55.0%   | 4.3     | 0.37   |
| Senegal | 22.2% | 24.4% | 25.6% | 27.8% | 46.6%   | 3.8     | -0.30  |
| Iraq    | 19.3% | 23.7% | 27.6% | 29.5% | 43.0%   | 3.7     | -0.34  |

- Most Likely Results:**
- France vs Senegal 1-1 (12.2%)
  - France vs Norway 1-1 (13.2%)
  - France vs Iraq 1-1 (13.2%)
  - Senegal vs Norway 0-0 (15.8%)
  - Senegal vs Iraq 0-0 (16.9%)
  - Norway vs Iraq 0-0 (22.1%)

GROUP J

| Team      | 1st   | 2nd   | 3rd   | 4th   | Qualify | Avg Pts | Avg GD |
|-----------|-------|-------|-------|-------|---------|---------|--------|
| Austria   | 30.4% | 27.9% | 23.8% | 17.8% | 58.4%   | 4.4     | 0.51   |
| Argentina | 30.1% | 25.0% | 22.7% | 22.3% | 55.1%   | 4.2     | 0.18   |
| Algeria   | 23.0% | 25.2% | 26.0% | 25.8% | 48.2%   | 3.9     | -0.08  |
| Jordan    | 16.5% | 21.8% | 27.5% | 34.2% | 38.3%   | 3.5     | -0.62  |

- Most Likely Results:**
- Argentina vs Austria 0-1 (14.9%)
  - Argentina vs Algeria 1-1 (13.4%)
  - Argentina vs Jordan 1-0 (13.8%)
  - Austria vs Algeria 0-0 (26.1%)
  - Austria vs Jordan 0-0 (27.2%)
  - Algeria vs Jordan 0-0 (22.1%)

GROUP K

| Team       | 1st   | 2nd   | 3rd   | 4th   | Qualify | Avg Pts | Avg GD |
|------------|-------|-------|-------|-------|---------|---------|--------|
| DR Congo   | 31.4% | 27.5% | 23.9% | 17.2% | 59.0%   | 4.6     | 0.71   |
| Uzbekistan | 26.5% | 26.6% | 25.1% | 21.8% | 53.1%   | 4.2     | 0.26   |
| Portugal   | 25.0% | 24.6% | 24.8% | 25.6% | 49.6%   | 3.9     | -0.13  |
| Colombia   | 17.1% | 21.2% | 26.2% | 35.4% | 38.3%   | 3.4     | -0.85  |

- Most Likely Results:**
- Portugal vs Colombia 1-1 (12.7%)
  - Portugal vs Uzbekistan 0-1 (13.6%)
  - Portugal vs DR Congo 0-1 (16.4%)
  - Colombia vs Uzbekistan 0-1 (16.2%)
  - Colombia vs DR Congo 0-1 (18.7%)
  - Uzbekistan vs DR Congo 0-0 (24.7%)

GROUP L

| Team    | 1st   | 2nd   | 3rd   | 4th   | Qualify | Avg Pts | Avg GD |
|---------|-------|-------|-------|-------|---------|---------|--------|
| England | 33.4% | 26.3% | 21.6% | 18.8% | 59.6%   | 4.5     | 0.48   |
| Croatia | 29.8% | 27.1% | 23.4% | 19.7% | 56.9%   | 4.3     | 0.36   |
| Ghana   | 24.3% | 25.8% | 25.2% | 24.8% | 50.1%   | 4.0     | -0.08  |
| Panama  | 12.6% | 20.8% | 29.8% | 36.8% | 33.4%   | 3.2     | -0.76  |

- Most Likely Results:**
- England vs Croatia 1-1 (13.4%)
  - England vs Ghana 1-1 (13.1%)
  - England vs Panama 0-0 (21.9%)

Croatia vs Ghana 1-0 (13.7%)  
Croatia vs Panama 0-0 (27.3%)  
Ghana vs Panama 0-0 (25.9%)

### 3. Tournament Simulation (100,000 runs)

Tournament simulation error: TournamentSimulator.\_\_init\_\_() got an unexpected keyword argument 'group\_simulator'